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Do mutual funds care about corporate governance?

The Chinese evidence

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Abstract

In the context of socially responsible investment, this study explored the role that corporate governance issues played in mutual funds' investment decision-making within the Chinese equity market. Considering 1 094 listed firms from 2008 to 2013, a period which followed the Non-tradable Share Reform in China, we find that mutual funds tilt towards investing in companies with good corporate governance. Furthermore, among the corporate governance mechanisms, ownership structure and information disclosure seem to be important determinants of institutional stock-holding decisions while less attention is paid to the board of directors. The results are robust to different estimation methods.

Key words

Mutual Fund; Stock-holding Preference; Corporate Governance; Socially Responsible Investment

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1. Introduction

Resulting from the increased tendency towards the institutionalization of equity markets, institutional investors are already a dominant force in mature markets as reflected by the magnitude of their shareholdings. For instance, 68% of the total equity market capitalization in the US were owned by institutional investors in 2007 (Lewellen, 2011); in the UK, Çelik and Isaksson (2014) suggest that only 10% of shares were still held by physical persons in 2012. The increasingly important role that institutional investors are playing in equity markets is also linked to greater responsibility towards the individuals who contribute their money to these institutions (Heineman and Davis, 2011), as well as a diverse group of other stakeholders (Aon, 2007). Given this context, it is therefore unsurprising to observe that a growing number of institutional investors (especially mutual funds) are embracing the concept of socially responsible investment (SRI) (Ballesterio, et al., 2012; Nofsinger and Varma, 2014). SRI is commonly considered as the investment process that “combines investors’ financial objectives with their concerns about environmental, social and governance (ESG) issues” (Eurosif, 2008). Currently, SRI is migrating from a fringe investment strategy to a mainstream policy (Marlowe, 2014; Sparkes and Cowton, 2004; Statman, 2006), especially in many developed markets¹. At the same time, SRI in emerging markets has gathered increasing attention in recent years (MERCER and IFC, 2009); it is unclear, however, if SRI strategy plays a similar role in institutional investors’ decision-making within the context of emerging markets compared to their compatriots in developed markets.

¹ According to the Forum for Sustainable and Responsible Investment (US SIF) 2014, there were USD 6.57 trillion or more assets managed under SRI strategy as of year-end 2013 in the US, representing that one out of every six dollars under management was invested according to SRI strategy. In the UK, £ 2.89 trillion assets were invested in the SRI market through various strategies by the end of 2013 (Eurosif, 2014). In addition, the SRI market is developing rapidly in Australia as well, according to the Responsible Investment Association Australasia (RIAA), there were USD 25 billion and USD 153.1 billion under management in core responsible investment portfolio and board responsible investment portfolio respectively by the end 2013.

China, the largest emerging market in the world, has experienced enormous changes in the investor structure of its equity market since the 1990s. Mutual funds (referred to as securities investment funds in China) represent the largest institutional investors in the Chinese equity market, and have been aware of the significance of SRI (China Social Investment Forum, China SIF, 2014) with mutual fund managers acknowledging that ESG issues cannot be ignored (MERCER and IFC, 2009). The question whether these mutual funds have shifted from mere awareness of responsible investment towards actively pursuing it as an investment strategy, however, remains unanswered within the Chinese context. Addressing this question should shed some light on whether or not mutual funds act proactively by incorporating ESG issues into their investment decision making. Against this background, the present study thus considered the G-factor of the ESG issues to assess mutual funds' investment attitudes towards corporate governance issues by focusing on the Chinese investment environment. In addition, this study was conducted during the period following the implementation of the Non-tradable Shares Reform² (the Reform) by considering 1 094 companies that were listed on the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE) over the period 2008 to 2013. The study adopted a factor analysis methodology to construct a corporate governance score in accordance with the Code of Corporate Governance for Listed Companies in China (henceforth the 2002 Code) and other related regulations issued by the China Securities Regulatory Commission (CSRC). Using the constructed corporate governance score, we examined the relationship between mutual fund ownership and the score by controlling for a variety of firm-level characteristics. We found that in general, mutual funds are likely to integrate corporate governance issues in their stock-holding decisions. The results remain robust when applying the two-stage least squares (2SLS) method, as well as when annual changes in the variables were considered. This study contributes to the existing literature in two ways: first by documenting the role of corporate governance issues in mutual funds' decision making within the Chinese context, and secondly by providing additional evidence on the way that mutual funds

² The Non-tradable Share Reform (the Reform) started in 2005, and it provided the solution to the historical issue where shares in the same company had different rights for public trading. As a result of the Reform, the Chinese stock market gained market liquidity, which increased the value of investing in the Chinese stock market and provided wider development opportunities for institutional investors.

approach their fiduciary duties.

The rest of this study is organized as follows. Section 2 introduces basic background information about mutual funds in China; Section 3 provides a review of the existing literature with regard to mutual funds' decision making in terms of corporate governance; Section 4 describes the data and methodology; Section 5 presents and discusses the study's regression results, and Section 6 concludes.

2. Mutual funds in China

In China, mutual funds developed rapidly since the first open-ended fund was launched in 2001. For instance, a total of USD 339 billion of assets were under management by mutual funds in 2011, representing an increase of 142% compared to the value of USD 10 billion reported in 2001³. In addition, even though mutual funds account for relatively high levels of stock market participation compared to the other types of institutional investors in China (e.g. social security funds, insurance companies, securities companies, qualified foreign institutional investors)⁴, their combined stockholding still represent a relatively small portion of the overall stock market capitalization. Furthermore, in order to improve the investor structure of capital markets, as well as to curb market speculation and stabilize stocks prices, Chinese authorities have implemented a series of actions to promote mutual funds' development. One example is the release of the Law of the People's Republic of China on Securities Investment Funds (henceforth the Fund Law) in 2004, which was revised in 2012. The Fund Law reinforces the fiduciary duty of fund managers, emphasizes the importance to protect individual investors and promotes the innovation of supervision mechanisms. As a result, the Fund Law consolidates the foundation for mutual funds' healthy development.

SRI in emerging markets has gained momentum in recent years, which can be partly attributed to increased concerns regarding corporate social responsibility (CSR) in

³ See http://www.cs.com.cn/tzjj/jjts/201206/t20120615_3371875.html

⁴ See http://www.ifc.org/wps/wcm/connect/ad66c38048855312ad9cff6a6515bb18/IFC_Breif_China_online.pdf?MOD=AJPERES&CACHEID=ad66c38048855312ad9cff6a6515bb18

stock markets (Managi *et al.*, 2012; Ortas *et al.*, 2012), as well as the development of SRI equity indexes (Muzindutsi and Sekhampu, 2013; Ortas *et al.*, 2012). In the Chinese context, CSR practices are developing and to some extent have achieved certain levels of success, which can be observed by the growing number of CSR reports being released, as well as increased regulation issued by the Chinese authorities (Zhang, 2014).

Regarding the development of SRI Indexes, the Taida Environmental Index (including 40 companies listed on either the SSE or the SZSE) that was released in 2008 is considered as the first themed SRI index focusing on Chinese companies. In 2009, the SSE and the SZSE also launched the SSE-SRI and the SHSZ-SRI indexes respectively. During the same year, the CNINFO-CBN-AEGON Industrial CSR Index (henceforth referred to as the CBN-AEGON index) was released, containing listed companies from both the SSE and the SZSE. In addition, some carbon-related SRI indexes, such as the CNINFO LOW-CARBON 50 index, can be found.

In line with the considerable attention that CSR has been receiving and the development of Chinese SRI indexes, SRI is evolving as an exciting new topic for investors in China. Although SRI practices in China started relatively late compared to other markets, it was followed by rapid growth. In terms of responsible investment, Chinese mutual funds have set a good example for other institutional investors, no matter in their scale and scope. In 2006, the Bank of China (BOC) Sustainable Growth Equity Fund, which could be considered as the first quasi-SRI fund in China, was launched. The Industrial Social Responsibility Securities Investment Fund, founded in 2008 by the AEGON-Industrial Fund Management Company (henceforth referred to as the AEGON-Industrial Fund), could be considered as the first truly authentic Chinese SRI fund. This fund takes sustainability and ethics into consideration; besides economic performance, it aims to promote the fulfilment of enterprises' social responsibility as well. Following the AEGON-Industrial Fund, the China Construction Bank (CCB) Principal Asset Management Company and China Universal Asset Management also issued SRI funds in 2010 and 2011 respectively. In

2011 the first green equity fund, namely the Industrial Green Investment Securities Investment Fund, was launched by the AEGON-Industrial Fund; this fund focuses exclusively on green technology industries.

By the end of October 2014, a total of 18 Chinese mutual funds were being managed according to responsible investment approaches, with combined assets of over CNY 22 billion under management (China SIF 2014). Furthermore, the majority of these Chinese SRI funds provided impressive performances. For instance, the CBN-AEGON index managed to outperform the SHSZ300 index since its inception⁵. Given the background that China is undergoing an economic transformation, moving from a resource-based and labor-based economy to a knowledge-based economy, the China SIF (2014) proposed that fund managers are less likely to attribute assets to high energy consumption and high pollution industries; in future, investment in emerging industries that are beneficial to the China's economic transition would most probably receive priority.

3. Review of related literature

Investing in good corporate governance firms would be an effective way for mutual funds to address their fiduciary duties, engage in SRI strategy, and ultimately benefit stakeholders. The topic concerning mutual funds' investment decision making with regards to corporate governance, however, remains relatively unexplored (McCahery *et al.*, 2014). As stated by Shleifer and Vishny (1997), corporate governance deals with “the way in which suppliers of finance to corporations assure themselves of getting a return on their investment”. Investing in firms with good corporate governance help to reduce monitoring and exit costs for mutual funds, as well as to enhance corporate value and shareholder wealth (Beiner *et al.*, 2006; Cheung *et al.*, 2010; Drobetz *et al.*, 2004). As a consequence, corporate governance becomes a critical consideration for mutual funds when making investment decisions (McCahery *et al.*, 2014), and there is a high control premium for firms with good corporate governance (Giannetti and Simonov, 2006).

⁵ See <http://www.yicai.com/news/2014/01/3422015.html>

In particular, institutional investors appear to pay great attention to board independence (see McCahery *et al.*, 2014; Useem *et al.*, 1993), and are likely to invest in firms where the positions of CEO and chairperson are held by different individuals (Chiu and Monin, 2003); in this regard, mutual funds are no exception. In addition, a firm's ownership structure, such as its ownership concentration (Barucci and Falini, 2005; Hussain, 2000) and its internal ownership (Frank and Ghosh, 2012), is an important consideration when selecting stocks; most results on the extent that ownership structure attracts mutual funds, however, are mixed. In terms of information disclosure, another important component of corporate governance, previous research has shown that firms with more accessible information are more popular among investors. Bushee and Noe (2000) found that firms with higher Association for Investment Management and Research (AIMR) disclosure rankings attracted more institutional investors, which means that disclosure practices affect the level of institutional ownership (see also Tang and Hu, 2011). Bradshaw *et al.* (2004) observed that institutions invest more heavily in firms that employed one of the so-called Big 5 auditing companies (now the Big 4) than in firms that acquired the services of other auditing companies. Aggarwal *et al.* (2005) arrived at the same conclusion and reported that US mutual funds in emerging market equities allocated more assets to firms with strong accounting standards.

In addition, the subject of mutual funds' stock-holding preferences has been extensively explored within the context of developed markets. Limited Chinese evidence, however, exists in terms of this topic. Although most previous studies observed that institutional investors (mutual funds included) appear more likely to select well-governed firms (Gao and Zhang, 2008; Jiang, 2007; Tang and Hu, 2011), the majority of these studies considered only one single corporate governance mechanism in isolation; these studies furthermore investigated the period before the Reform. Hence, relatively little is known about mutual funds' preferences from both a board and systematic corporate governance perspective during the period following the implementation of the Reform.

4. Data and methods

4.1 Sample

The sample included in this study contained companies listed on the main boards of the SSE and the SZSE. The data pertaining to mutual fund ownership is collected from the RESSET database, while the remaining data is obtained from the CSMAR database.

To assess the role of corporate governance in mutual funds' decision making after the Reform (basically most activities related to the Reform were completed by 2007), the period covered by this study is from 2008 to 2013. Companies that are listed in the financial sector, as well as companies with missing data, less than one listing year by the end of 2008 and those that were specially treated (ST or *ST) are excluded from the study. After excluding companies based on the above-mentioned criteria, the sample included in this study consisted of 1 094 listed companies, providing 6 564 observations over the study period of 6 years.

4.2 The construction of the corporate governance score

Since Martindell (1950) provided an appraisal of the directorate for manufacturing companies, several scholars attempted to develop corporate governance assessment systems. Two well-known examples include Gompers *et al.* (2003) with the Governance Index, and Brown and Caylor (2004) with the Gov-Score index. At the same time, commercial organizations also launched a series of corporate governance evaluation indices. These include Standard & Poor's Corporate Governance Scores, Deminor Rating's Corporate Governance Ratings, and Credit Lyonnais Securities Asia's (CLSA) corporate governance index. It should be noted, though, that the application of these corporate governance indexes differ, essentially due to the heterogeneous corporate governance environment (e.g. regulations, laws, policy, development level of corporate governance that are applicable) and also the listed companies included in indices (The project team of the Research Center for Corporate Governance of Nankai University, 2003). Hence, it is questionable to generalize these

corporate governance indices and to directly apply them to companies from different corporate governance environments.

Given the development of Chinese capital markets since the 1990s, the corporate governance issue has increasingly drawn attention from academics and practitioners alike. In terms of corporate governance appraisal, the Research Center for the Studies of Corporate Governance at Nankai University is considered as the pioneer, and launched the China Corporate Governance Index (CCGI^{NK}) in 2004. This index consists of six first-level dimensions focusing on shareholders' rights, the board director and directorate, the supervisor and supervisory board, management, information disclosure and stakeholders. Chinese scholars, such as Jin and Yuan (2008), Liao *et al.* (2008) and Shen *et al.* (2009) also attempted to build corporate governance assessment systems, and their works are of considerable referential importance for the present study.

In this section, we discuss the selection of corporate governance indicators that are relevant in the Chinese context, and adopt a factor analysis methodology to construct a corporate governance score (CG_SCORE).

4.2.1 Indicator selection for the CG_SCORE

Based on the 2002 Code, Chinese Company Law (2014) and some regulations issued by the CSRC, as well as considering the data availability, the indicators selected for constructing the CG_SCORE are indicated as follows⁶.

Ownership structure Being the core mechanism of a corporate governance system (Kim *et al.*, 2004; Liang *et al.*, 2011), corporate governance is usually examined from two dimensions, namely, ownership concentration and ownership composition. In this study, the proportion of shares held by the largest shareholder (CR1) and the degree of balance provided by the second to tenth largest shareholders to the largest shareholder

⁶ Corporate governance mechanisms can generally be divided into internal and external mechanisms (Denis & McConnell, 2003). Since the listed companies included in this study are all facing a similar external governance environment, only internal mechanisms are taken into account.

(Z10) are employed as proxies of ownership concentration; ownership composition⁷ is represented by director ownership (DHOLD) and managerial ownership (MHOLD), which are viewed as internal ownership as well.

As reported in Table 4.1, the mean (median) of CR1 and Z10 is 35.9% (33.8%) and 61.4% (39.5%) respectively, to some extent indicating that listed companies are characterized by concentrated ownership. Table 4.1 reflects that the mean of MHOLD and DHOLD is 0.3% and 0.7% respectively, an indication that both directors and managers hold only a small fraction of the total ownership.

Board of directors⁸ A company's board of directors plays a monitoring role during decision-making and risk management, and act as a safeguard for the corporate governance mechanism. This study includes board size (DSIZE), director independence (INDEP), CEO duality (DUAL), and the number of committees (COMM) to represent the characteristics of the board of directors.

In Table 4.1, the mean of DSIZE is more or less 9, close to the median of 9. The mean (median) of INDEP is 36.7% (33.3%), which is consistent with the mandatory requirement indicated in the Establishment of Independent Director Systems by Listed Companies Guiding Opinion (2001) that independent directors should at least account for one-third of the total board members. In addition, the mean of DUAL is 0.128, indicating that in most listed companies, the positions of manager and chairperson are held by different persons. Furthermore, the average number of committees is 4, in accordance with the 2002 Code that recommends that companies should establish a corporate strategy committee, audit committee, nomination committee and remuneration and appraisal committee.

⁷ In previous studies concerning this topic, tradable shareholding is frequently taken into consideration. With the Reform basically being completed before the study period of this study commenced, most shares have become tradable, thus tradable shareholding is not a great concern in this study.

⁸ Chinese listed companies feature a two-tier board system, that is, listed companies should establish a board of directors, as well as a supervisory board. However, in most previous studies (e.g. Hu *et al.*, 2010; Peng *et al.*, 2007), it was found that the supervisory board has no significant effect on corporate governance. The regulation on board independence released by the CSRC in 2001 is targeted at the board of directors only. For this reason, the supervisory board is not included in the CG_SCORE.

Information disclosure Information disclosure and transparency are major drivers of good corporate governance (Munisi *et al.*, 2014). This study selected a firm's auditor (AUDIT) and cross listing (CROSS) for the measurement of information disclosure. In Table 4.1, the mean and median values for AUDIT and CROSS is low, indicating that only a small number of companies employed one of the Big 4 as their auditor, and that only a few of them are cross listed on other stock exchanges.

Table 4.1 Descriptive statistics on selected data considered for corporate governance score construction

Variable	Obs.	Mean	Std. Dev.	Min	Max	Med
CR1	6 564	0.359	0.158	0.022	0.864	0.338
Z10	6 564	0.614	0.647	0.008	6.686	0.395
MHOLD	6 564	0.003	0.022	0.000	0.396	0.000
DHOLD	6 564	0.007	0.041	0.000	0.493	0.000
INDEP	6 564	0.367	0.054	0.091	0.714	0.333
DSIZE	6 564	9.207	1.958	4.000	18.000	9.000
COMM	6 564	3.892	0.591	0.000	8.000	4.000
DUAL	6 564	0.128	0.334	0.000	1.000	0.000
AUDIT	6 564	0.074	0.261	0.000	1.000	0.000
CROSS	6 564	0.037	0.190	0.000	1.000	0.000

4.2.2 CG_SCORE formulation

A factor analysis methodology is employed to formulate the CG_SCORE. The KMO value from the Kaiser-Meyer-Olkin (KMO) test for this study is larger than 0.5, indicating that the data is acceptable for factor analysis (Kaiser, 1974). Also, Bartlett's test is significant at the 1% level, thereby rejecting the null hypothesis that the correlation matrix is an identity matrix and suggesting that factor analysis is appropriate.

Table 4.2 shows the initial eigenvalues, as well as eigenvalues after the extraction and rotation corresponding to each component. Five factors are left with eigenvalues larger than 1, which are included as principal factors for the corporate governance score formulation. The percentage of cumulative variance explained after rotation is 72.17%, suggesting that the five factors are likely to reflect the level of corporate governance comprehensively.

Table 4.2 Total variance explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.044	20.442	20.442	2.044	20.442	20.442	1.746	17.457	17.457
2	1.550	15.497	35.939	1.550	15.497	35.939	1.632	16.317	33.774
3	1.384	13.844	49.783	1.384	13.844	49.783	1.492	14.923	48.697
4	1.213	12.134	61.918	1.213	12.134	61.918	1.317	13.169	61.866
5	1.026	10.257	72.174	1.026	10.257	72.174	1.031	10.308	72.174
6	0.911	9.107	81.282						
7	0.658	6.580	87.862						
8	0.583	5.826	93.688						
9	0.344	3.437	97.125						
10	0.288	2.875	100.000						

Extraction Method: Principal Component Analysis.

The rotated component matrix is displayed in Table 4.3, and indicates the factor loadings for each variable onto each factor; loadings smaller than 0.4 is not shown in Table 4.3. Therefore, five factors are extracted with factor analysis. Factor 1 to factor 5 are designated by the terms CG_1 to CG_5, respectively. CG_1 reflects internal ownership, the high CG_1 signifies high MHOLD and DHOLD. In a similar fashion, CG_2 represents ownership concentration, with the high CG_2 value reflecting low CR1 and high Z10; CG_3 portrays information disclosure, with a high CG_3 value suggesting a high possibility of a company employing a Big 4 auditor and being listed on other stock exchanges. CG_4 and CG_5 jointly represent the board of directors. CG_4 reflects board independence, with a high CG_4 indicating high INDEP and small DSIZE. CG_5 combines COMM and DUAL, where a high CG_5 signifies that there are more special committees established under the board of directors and a high possibility of different individuals serving as the chairperson and the CEO.

Table 4.3 Rotated Component Matrix ^a

Variable	Component				
	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
MHOLD	0.918				
DHOLD	0.906				
CR1		-0.896			
Z10		0.882			
AUDIT			0.824		

CROSS	0.818		
INDEP		0.811	
DSIZE		-0.752	
COMM			0.900
DUAL			-0.403

Extraction Method: Principal Component Analysis; Rotation Method: Varimax with Kaiser Normalization.

^a. Rotation converged in five iterations.

Based on the extracted five factors, this study adopted the weighted objectives method, and took the contribution rate of each factor as weight and used these weightings to construct the CG_SCORE, as reflected by Equation 4.1.

$$CG_SCORE = 0.241874 * CG_1 + 0.226079 * CG_2 + 0.206764 * CG_3 + 0.182462 * CG_4 + 0.142822 * CG_5 \quad (4.1)$$

4.3 Variables

Dependent variable To explore mutual funds' preferences for corporate governance, this study used the level of ownership in a company by mutual funds (MF) to represent their preferences towards the company. The same measurement is employed in studies by Bennett *et al.* (2003) and Bushee *et al.* (2014), with a high level of MF reflecting greater mutual funds preference towards an investee company.

Table 4.4 reports the stockholding distribution of mutual funds over the study period. The mean (median) of MF is shown as 5.2% (1.3%) suggesting that mutual funds account for a relatively small portion of the stock market in terms of their shareholding; their average shareholding decreased from 6.5% to 3.5% during the period 2009-2013, partly due to the negative effect of the 2007-2008 global financial crisis on the stock market.

Table 4.4 Descriptive statistics on mutual fund ownership of the sample

Year	Obs.	Mean	Std. Dev.	Min	Max	Med
2008	1 094	0.057	0.101	0.000	0.665	0.006
2009	1 094	0.065	0.091	0.000	0.544	0.021
2010	1 094	0.061	0.087	0.000	0.543	0.025
2011	1 094	0.052	0.082	0.000	0.519	0.017
2012	1 094	0.043	0.075	0.000	0.501	0.011
2013	1 094	0.035	0.064	0.000	0.407	0.006

2008-2013	6 564	0.052	0.085	0.000	0.665	0.013
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Independent variable This study assessed the role that corporate governance played in the investment decision-making of mutual funds. The CG_SCORE constructed in Section 4.2 was used as the proxy for corporate governance. As indicated in Table 4.5, the average CG_SCORE value increased consistently from -0.047 in 2008 to 0.049 in 2013, generally indicating that the sample companies are doing relatively well in improving corporate governance practices, and also signifying that corporate governance reform is proving to be effective. The independent variable included five individual corporate governance factors (i.e. CG_1, CG_2, CG_3, CG_4, and CG_5) to allow for a more detailed analysis of the relationship between corporate governance and mutual funds.

Table 4.5 Descriptive statistics on the corporate governance score

Year	Obs.	Mean	Std. Dev.	Min	Max	Med
2008	1 094	-0.047	0.458	-1.175	2.882	-0.107
2009	1 094	-0.032	0.437	-1.336	2.971	-0.099
2010	1 094	-0.003	0.427	-1.192	2.481	-0.077
2011	1 094	0.015	0.433	-1.192	3.354	-0.067
2012	1 094	0.021	0.486	-1.191	3.233	-0.070
2013	1 094	0.049	0.475	-1.199	3.241	-0.033
2008-2013	6 564	0.000	0.454	-1.336	3.354	-0.079

Controlling variables Based on previous studies (e.g. Badrinath *et al.*, 1989; Brands *et al.*, 2006; Covrig *et al.*, 2006; Ferreira and Matos, 2008; Grinstein and Michaely, 2005; Lai *et al.*, 2013; Pinnuck, 2004), particularly those studies concerning fiduciary duties as prescribed by the prudent man rules, this study employed some firm-level characteristics as controlling variables. The controlling variables included are return on assets (ROA), reflecting the firm's operating performance; stock return volatility (VOL), beta (BETA) and financial leverage (LEV), reflecting share risk; the share turnover ratio (TURN) as an indicator of liquidity; firm size (SIZE), listing history (AGE), dividend payout ratio (DP) and the SHSZ300 index as components of institutional investment constraints. The descriptive statistics of the controlling variables are reported in Table 4.6.

Table 4.6 Descriptive statistics on firm-level characteristics

	Obs.	Mean	Std. Dev.	Min	Max	Med
ROA	6 564	0.037	0.060	-0.722	0.477	0.031
DP	6 564	0.646	3.082	0.000	41.667	0.294
LEV	6 564	0.522	0.192	0.002	1.094	0.537
SIZE	6 564	22.105	1.345	17.467	28.482	21.975
AGE	6 564	2.449	0.395	0.000	3.135	2.485
VOL	6 564	0.471	0.264	0.125	18.428	0.440
TURN	6 564	0.392	0.288	0.002	2.586	0.322
BETA	6 564	0.999	0.194	-3.415	1.765	1.022
SHSZ300	6 564	0.193	0.394	0.000	1.000	0.000

In order to examine the robustness of the results, this study explored the influence of one-year changes in corporate governance (CCG_SCORE) on changes in mutual fund ownership (CMF). Thus, the variables CMF, CCG_SCORE, as well as the changes in firm-level characteristic variables (i.e. CROA, CLEV, CSIZE, CAGE, CVOL, CTURN, CBETA, CDO, CSHSZ300) are also employed as dependent variables, independent variable, and controlling variables respectively.

4.4 Models

Based on the analysis above, Model 4.2 was constructed to examine whether mutual funds care about corporate governance when they select stocks. In terms of the endogeneity problem, both ordinary least squares (OLS) and 2SLS regression estimations were used to avoid potentially biased results.

$$MF_{it} = \alpha_0 + \beta_1 CCG_SCORE_{it} + \lambda_1 ROA_{it} + \lambda_2 LEV_{it} + \lambda_3 SIZE_{it} + \lambda_4 AGE_{it} + \lambda_5 VOL_{it} + \lambda_6 TURN_{it} + \lambda_7 BETA_{it} + \lambda_8 DP_{it} + \lambda_9 SHSZ300_{it} + \varepsilon_{it} \quad (4.2)$$

In order to further explore the relationship between mutual funds' stock-holding preferences and corporate governance, changes in corporate governance and changes in mutual fund ownership are employed in Model 4.3. If corporate governance is a significant driver in mutual funds' investment decision-making, changes in mutual fund ownership is expected to increase with changes in corporate governance.

$$MF_{it} = \alpha_0 + \beta_1 CCG_SCORE_{it} + \lambda_1 CROA_{it} + \lambda_2 CLEV_{it} + \lambda_3 CSIZE_{it} + \lambda_4 CAGE_{it} + \lambda_5 CVOL_{it} + \lambda_6 CTURN_{it} + \lambda_7 CBETA_{it} + \lambda_8 CDP_{it} + \lambda_9 CSHSZ300_{it} + \varepsilon_{it} \quad (4.3)$$

5. Results and discussion

5.1 Corporate governance issues considered during stock-holding decision making

This section explored mutual funds' stock-holding preferences for corporate governance. Table 5.1 reports the relationship between mutual fund ownership and individual corporate governance factors (CG_1 to CG_5).

In Table 5.1, CG_1 exhibits a negative relationship with MF (significant at the 1% level), indicating that mutual funds tilt towards investing in firms with low levels of internal ownership. This result is inconsistent with prior studies (e.g. Jensen and Meckling, 1976; Kim *et al.*, 2004) where internal ownership (or management ownership) is considered as a good way for managers to align with shareholders, and helps to alleviate agency problems between them. As shown in Table 4.1, director ownership and manager ownership is still very low in China. According to the Chinese evidence from Wang and Xu (2009), if the proportion of managerial shareholding is too small, it is unlikely to mitigate the conflicts between managers and owners; some managers moreover breach their fiduciary duties. In such a situation, managers have no incentives to maximize corporate performance; conversely, it provides opportunities to pursue benefits for themselves. Additionally, Chinese companies are characterized by concentrated ownership, and ownership incentives are not able to function properly in the presence of a controlling shareholder (Li and Li, 2006).

CG_2 is significantly and positively related with MF at the 1% level, suggesting that mutual funds prefer firms with a relatively low proportion of shares being held by the largest shareholder and a high degree of balance offered against the largest shareholder. In China, a well-known feature of listed companies is that they have concentrated ownership structures (Wang and Xiao, 2011; Ma *et al.*, 2010). Some researchers (e.g. Demsetz and Villalonga, 2001; La Porta *et al.*, 2000; Miller *et al.*, 2007; Shleifer and Vishny, 1997) suggest that highly concentrated ownership provides power and incentives for dominant shareholders to more easily pursue their own

Table 5.1 Regression results on the relationships between mutual fund ownership and individual corporate governance factors

This table presents the results of the relationships between MF and CG_1 to CG_5 for a sample of 1 094 listed firms over the period 2008 to 2013. A fixed effect model is adopted in all regressions with the F test and Hausman test rejecting the null hypothesis. We control for year and industry fixed effects in all regressions. Variable definitions are reported in Appendix A. T-statistics are reported in parentheses; ***, **, * represent significance at 1%, 5% and 10%, respectively.

	1		2		3		4		5	
	Coefficient	Std. Error	Coefficient	Std. Error	Coefficient	Std. Error	Coefficient	Std. Error	Coefficient	Std. Error
Intercept	0.707*** (4.80)	0.147	0.750*** (5.23)	0.144	0.734*** (5.01)	0.092	0.727*** (4.97)	0.146	0.724*** (4.88)	0.148
CG_1	-0.013*** (-2.68)	0.005								
CG_2			0.037*** (6.83)	0.005						
CG_3					0.011** (2.27)	0.004				
CG_4							0.001 (0.21)	0.002		
CG_5									-0.001 (-0.25)	0.002
ROA	0.492*** (9.49)	0.052	0.495*** (9.64)	0.051	0.492*** (9.49)	0.037	0.490*** (9.47)	0.052	0.489*** (9.45)	0.052
LEV	0.090*** (4.54)	0.020	0.113*** (5.61)	0.020	0.092*** (4.60)	0.017	0.090*** (4.47)	0.020	0.089*** (4.46)	0.020
SIZE	-0.028*** (-3.91)	0.007	-0.030*** (-4.34)	0.007	-0.029*** (-4.11)	0.004	-0.029*** (-4.05)	0.007	-0.028*** (-3.97)	0.007

AGE	-0.020** (-2.04)	0.010	-0.020** (-2.05)	0.010	-0.020** (-2.01)	0.009	-0.020** (-2.05)	0.010	-0.020** (-2.06)	0.010
VOL	0.057*** (4.01)	0.014	0.052*** (3.72)	0.014	0.056*** (3.91)	0.013	0.056*** (3.91)	0.014	0.056*** (3.90)	0.014
TURN	-0.026*** (-4.09)	0.006	-0.022*** (-3.57)	0.006	-0.025*** (-3.96)	0.006	-0.026*** (-4.01)	0.006	-0.026*** (-4.01)	0.006
BETA	-0.057*** (-5.61)	0.011	-0.052*** (-5.18)	0.010	-0.056*** (-5.62)	0.009	-0.056*** (-5.54)	0.010	-0.056*** (-5.54)	0.010
DP	-0.000 (-1.26)	0.000	-0.000 (-1.33)	0.000	-0.000 (-1.28)	0.000	-0.000 (-1.27)	0.000	-0.000 (-1.25)	0.000
SHSZ300	0.015*** (2.83)	0.005	0.014*** (2.69)	0.005	0.015*** (2.80)	0.005	0.015*** (2.78)	0.005	0.015*** (2.79)	0.005
<i>N</i>	6 564		6 564		6 564		6 564		6 564	
<i>Adj. R²(%)</i>	76.62		77.45		76.59		76.53		76.53	

wealth at the expense of minority shareholders; this could result in tunneling (Bai *et al.*, 2004; Johnson *et al.*, 2000) and hence decreases corporate profitability (Gomez *et al.*, 2001). In addition, as mentioned by Bennedsen and Wolfenzon (2000) and Edmans and Manso (2011), the presence of several large shareholders might provide more efficient monitoring than only one single majority shareholder.

Furthermore, mutual funds appeared to be interested in firms that are audited by the Big 4 auditor firms and that are cross listed on other stock exchanges, since CG_3 has a positive relationship with MF (significant at the 5% level). This finding is consistent with the majority of prior literature where it is reported that firms with more accessible information are more popular among investors (as mentioned in Section 3). Legal protection systems for investors are relatively poor in China, and the presence of controlling shareholders might also reduce information transparency (Attig *et al.*, 2006), which compels investors to avoid stocks with limited information.

However, the board of directors seemed not to be a key consideration for mutual funds' investment, since neither CG_4 nor CG_5 were significantly associated with MF. One possible reason is that the attributes of the board of directors may not seem to be significantly associated with corporate performance (Cornett *et al.*, 2007; Erickson *et al.*, 2005; Rose, 2005). In addition, many aspects relating to the board of directors have already been set according to the 2002 Code (as seen in Table 4.1); thus there is no significant difference among listed companies in this regard. Besides, Chinese scholars (e.g. Shao *et al.*, 2004; Tang *et al.*, 2013) find that with the increasing shareholdings held by the largest shareholders, they are more easily able to dominate the board of directors and turn the board of directors into a channel for them to expropriate firm wealth. If so, the board of directors seems more symbolic and lacks its intended monitoring function. Therefore, mutual funds might doubt the role of the board of directors within the concentrated ownership structures of listed companies in the Chinese stock market.

We then continued by placing greater emphasis on examining the relationship between the overall corporate governance score and mutual funds' stock-holding

preferences; the results are represented in Table 5.2. Mutual funds appear to prefer firms with good corporate governance as reflected by the evidence reported in Column (1) of Table 5.2, where it is indicated that MF is positively and significantly associated with the CG_SCORE. The result shows that mutual funds generally are in favor of firms with good corporate governance, suggesting that mutual funds incorporate corporate governance issues as part of their investment decision making; it also provides indirect evidence that mutual funds believe that good corporate governance is beneficial to mitigate monitoring costs, as well as to enhance shareholder value. It is furthermore associated with governance reforms that encourage listed companies to develop corporate governance mechanisms and compel institutional investors to have a thorough understanding of corporate governance (Chen and Zhang, 2014). Accordingly, the governance reforms in China appear to have succeeded. Responsible investment for mutual funds appears to have shifted from awareness to action. The results are furthermore robust when using 2SLS regression, as can be seen in Column (2) of Table 5.2.

Table 5.2 Regression results on the relationships between mutual fund ownership and corporate governance

This table presents the results of the relationships between MF and the CG_SCORE for a sample of 1 094 listed firms over the period 2008 to 2013. Column 1 shows OLS regression results from a fixed effect model; Column 2 shows 2SLS regression results by employing log of number of shareholders as the instrument variable of CG_SCORE, the null hypothesis of a weak instrument is rejected based on the value of the Cragg–Donald F-statistic being larger than the critical value provided by Stock and Yogo (2005), and the Hausman test for the endogeneity problem where the null hypothesis is not rejected testifies that CG_SCORE is exogenous. We control for year and industry fixed effects in all regressions. Variable definitions are reported in Appendix A. T-statistics are reported in parentheses; ***, **, * represent significance at 1%, 5% and 10%, respectively.

	OLS		2SLS	
	1		2	
	Coefficient	Std. Error	Coefficient	Std. Error
Intercept	0.763*** (5.22)	0.146	0.895*** (8.75)	0.102
CG_SCORE	0.023*** (2.94)	0.008	0.107*** (4.63)	0.023
ROA	0.492*** (9.56)	0.051	0.500*** (13.15)	0.038
LEV	0.094***	0.020	0.110***	0.018

	(4.67)		(6.07)	
SIZE	-0.030*** (-4.31)	0.007	-0.037*** (-7.37)	0.005
AGE	-0.020** (-1.98)	0.010	-0.019** (-2.01)	0.009
VOL	0.057*** (3.98)	0.014	0.062*** (4.51)	0.014
TURN	-0.025*** (-3.92)	0.006	-0.022*** (-3.61)	0.006
BETA	-0.056*** (-5.57)	0.010	-0.056*** (-6.07)	0.009
DP	-0.001 (-1.38)	0.000	-0.001 (-1.63)	0.000
SHSZ300	0.015*** (2.72)	0.005	0.013*** (2.65)	0.005
N	6 564		6 564	
Adj. R ² (%)	76.64		12.01	

The regressions results obtained when investigating the relationships between changes in corporate governance and changes in mutual ownership are reported in Table 5.3. CMF is positively and significantly related with the CCG_SCORE, indicating that changes in mutual fund ownership are in line with changes in corporate governance. The results moreover indicate that the stock-holding preferences of mutual funds towards corporate governance are robust.

Table 5.3 Regression results on the relationships between changes in mutual fund ownership and changes in corporate governance

This table presents the results of the relationships between CCG_SCORE and CMF for a sample of 1 094 listed firms. Due to one-year changes in variables being considered, 5 470 observations from 2009 to 2013 are included in the regressions. We control for year and industry effects in all regressions. Variables included in the regressions are defined according to Appendix A. T-statistics are reported in parentheses; ***, **, * represent significance at 1%, 5% and 10%, respectively.

	Coefficient	Std. Error
Intercept	-0.020*** (-4.91)	0.004
CCG_SCORE	0.023*** (2.61)	0.009
CROA	0.120** (2.40)	0.050
CLEV	0.068**	0.029

	(2.32)	
CSIZE	-0.011 (-1.03)	0.011
CAGE	0.127*** (3.19)	0.040
CVOL	0.033* (1.84)	0.018
CTURN	-0.010 (-1.48)	0.007
CBETA	-0.029** (-2.47)	0.012
CDP	-0.002** (-1.99)	0.001
CSHSZ300	-0.004 (-0.62)	0.006
<i>N</i>		5 470
<i>Adj. R²(%)</i>		21.48

5.2 Mutual funds' preferences for other firm-level variables

In Tables 5.1 and 5.2, the positive (negative) relationships observed for MF with LEV and VOL (SIZE, AGE and TURN) illustrate that mutual funds do not obey prudent man standards. As expected however, MF is significantly and positively related with ROA and SHSZ300 (an indication that mutual funds are attracted by firms with healthy financial performance and SHSZ300 constituent firms). When considering the relationship between changes in controlling variables and mutual fund ownership, some results shown in Table 5.3 differ somewhat from the previous; the majority of the results, however, are similar to those reported in Tables 5.1 and 5.2.

With regard to other firm-level characteristics, mutual funds do not always seem to be subject to prudent man rules, which are inconsistent with the situation reported in developed markets. Irrational behavior by mutual funds in China could partly be ascribed to the immature stock market and undeveloped institutional environment. To better meet their fiduciary duties, mutual funds (who are still at the early stages of their development in China) as well as Chinese authorities therefore still have a long way to go.

6. Conclusion

This study adopted a factor analysis method to formulate a corporate governance score (CG_SCORE). By considering the CG_SCORE, we explored the role that corporate governance issues played in mutual funds' investment decision-making. Regression results show that in general, mutual funds appear to care about corporate governance issues. . In particular, mutual funds place greater emphasis on ownership structure and information disclosure, and less on the board of directors. The relationship remains robust when applying 2SLS and changes in the value of the variables. In this regard, mutual funds in China appear to have engaged in responsible investment.

Nonetheless, SRI has not been incorporated into mainstream mutual fund investment in China, and only a small portion of mutual funds are SRI driven. The results of this study provide some implications for mutual funds to further approach responsible investment, as well as for regulators, policy makers, and academics to conduct further research on this topic.

To begin with, mutual funds are still relatively small in size compared to the overall equity market; expanding the scale of mutual fund ownership in the Chinese equity market could be the premise for future SRI market development. Secondly and even more importantly, the governance of mutual funds presents a great concern that should receive attention; badly governed mutual funds are unlikely to effectively engage in SRI. Effective regulation and supervision should be implemented to assist mutual funds to improve their internal governance, as well as their own CSR performance. Thirdly, SRI training and education seem necessary for investors and fund managers since their understanding of SRI still remains at a theoretical level to a certain extent. Fourthly, CSR and SRI indexes are the prominent factors influencing investors' responsible investment. Thus it seems necessary to examine the practicality of CSR reporting as well as the reliability of a SRI index, to help mutual funds and other investors save SRI screening cost, and provide reliable information for referencing. Last but not least, it should be noted that sufficient risk assessment

cannot be ignored. It would be meaningless to discuss advancing social good without taking risk and financial factors into consideration.

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Appendix A

Variable definitions

Variable	Definition
IO_TOTAL	Proportion of shares held by mutual fund
CG_SCORE	Corporate governance score
CG_1	Corporate governance factor representing MHOLD and DHOLD
CG_2	Corporate governance factor representing CR1 and Z10
CG_3	Corporate governance factor representing AUDIT and CROSS
CG_4	Corporate governance factor representing DSIZE and INDEP
CG_5	Corporate governance factor representing COMM and DUAL
CR1	Proportion of shares held by the largest shareholder
Z10	Proportion of shares held by the second to tenth largest shareholders / CR1
MHOLD	Proportion of shares held by managers
DHOLD	Proportion of shares held by directors
DSIZE	Number of directors serving on the board
INDEP	Number of independent directors / DSIZE
COMM	Number of special committees under the board of directors
DUAL	If the positions of CEO and chairperson are held by different individuals, the value equals 1, otherwise 0
AUDIT	If the firm is audited by a Big 4 auditing firm, the value equals 1, otherwise 0
CROSS	If the firm is listed on other stock exchanges besides the SSE and the SZSE, the value equals 1, otherwise 0
ROA	Net income / average assets
DP	Dividends / net income
LEV	Total debt / total assets
SIZE	Log of total assets
AGE	Log of listing years
VOL	Standard deviation of daily stock returns over the preceding twelve months
TURN	Average monthly trading volume relative to total shares outstanding for the preceding twelve months
BETA	Calculated by means of a market model using daily stock returns for the preceding twelve months
SHSZ300	If the firm is a constituent of the SHSZ300 index, the value equals 1, otherwise 0